

FEEL BEFORE YOU BUILD

Hands-on prototyping with actuators · 2 hours

WHY WE ARE HERE

You can read a datasheet. You have never held one of these.

Most projects end up on the screen and the speaker

- Not because touch was wrong
- Because nobody knew a coin motor costs 12 kr

Today: decide it feels cheap, pick something else

- Better now than in week 46

HOW TODAY WORKS

Six benches, already wired and running

- Ten minutes each. Hard timer
- Change one parameter, answer one question
- You wire nothing today

Then build one signal

- Someone else has to read it

RUN SHEET

Times are offsets from the start. The rotation is the spine of the session, so keep it moving.

0:00	10 min	Two motors, same message. Which felt urgent?
0:10	30 min	Bench tour: touch all six, five minutes each
0:40	10 min	Round-the-room: what surprised you
0:50	50 min	Tasks. Grab a board, work at your own pace
1:40	15 min	Blind test on task 5
1:55	5 min	Pack down

THE SIX BENCHES

01 ERM COIN MOTOR

VIBRATION

An off-centre weight on a motor shaft.

The same part as in your phone.

Speed and strength are welded together.

"Gentle but fast" is not on offer.

USED FOR

phone notifications, game controller rumble,
anything worn under clothing.

part 10mm 3V coin ERM

drive 2N2222 + 1N4148

pin D9 (PWM)

draw ~75 mA

spin-up 20-40 ms

cost ~12 kr

CARD QUESTION

At what point does it stop feeling like information and start feeling like a malfunction?

02 LRA + PIEZO DISC

VIBRATION

Two ways out of bench 01's compromise.

LRA: resonance in 5 ms, stops just as fast.

Piezo: near-instant, almost no power.

A tick, not a thump.

USED FOR

keyboard clicks, Apple Watch taps, anything where lag would feel broken.

parts LRA 10mm + piezo

driver DRV2605L (I2C)

pin A4 SDA / A5 SCL

effects 123 waveforms

cost ~90 kr

CARD QUESTION

Which of the three vibrators would you actually put on a wrist, and why not the others?

03 SOLENOID TAP

IMPACT

One 15 ms pulse against a fingertip.

A knock reads as a person tapping you.

Buzzing reads as a machine.

CARD QUESTION

How many taps before it goes from alert to nagging?

USED FOR

navigation cues on the body, braille cells, alerts that must cut through noise.

part 5V push-pull

drive MOSFET + flyback

pin D6

peak ~1.1 A

duty <= 25%, gets hot

cost ~45 kr

04 CAPACITIVE TOUCH

TOUCH INPUT

Bare finger on a surface. Servos react.

No button. No sensor you can point at.

The input is a wire behind a plate.

CARD QUESTION

Put your hand near the plate without touching. When exactly did it decide that was a touch?

USED FOR

invisible controls in wood or fabric, waterproof panels, touch-sensitive prototypes.

sense ADCTouch on A0

baseline 25-sample roll

trigger 1.01x average

debounce 100 ms

servos D4 + D13

cost ~0 kr, a wire

BENCH 04: THE WHOLE TRICK

Touch input for the price of a wire.

reading > baselineAverage() * 1.01

Threshold is relative, never absolute

- Readings drift with humidity and mains hum
- Fixed threshold works at 9am, fails at 1pm

Baseline only learns while untouched

- Or a long press teaches it that a finger is normal

05 PELTIER WARM / COOL

THERMAL

Heats one side, cools the other.

Flip the current, it reverses.

Sit with it. Seconds before you are sure.

Hot alternating with cold reads as lukewarm.

USED FOR: slow ambient state, not alerts.

"The room is busy", never "you have mail".

part TEC1-12706
drive L298N H-bridge
pin D3 PWM / D4-5
draw 2-4 A, bench PSU
onset 3-8 s
cap 45 C in software

06 THE SAME SIGNAL, TWICE

AUDIO + TOUCH

One driver. 40 Hz you feel, 400 Hz you hear.

Together is not louder. It is more certain.

Two channels, same message.

CARD QUESTION

With ear defenders on, does the signal still work?

USED FOR

noisy or bright environments, and accessibility.
If one channel fails, one lands.

part bone-conduction

amp PAM8403 class-D

pin D11 (tone)

felt ~40 Hz

heard ~400 Hz

cost ~110 kr

NOW MAKE ONE DO SOMETHING

FIVE TASKS, ONE BOARD EACH

Upload, change two numbers, upload again. That loop is the point.

- 01 Make something spin**
- 02 Make something tap**
- 03 Make something warm**
- 04 Make it notice you**
- 05 Make it answer back**

1 to 3 are output. 4 is input. 5 is both.

You are not expected to finish all five.

HOW THE TASK SKETCHES WORK

Every sketch runs the moment you upload it. Nothing to wire.

A CHANGE ME block at the top

- Two or three numbers. Edit, upload, feel the difference

A THINGS TO TRY list at the bottom

- Four experiments, roughly in order of difficulty
- The last one is usually the interesting one



INPUTS, AND PRIOR ART

INPUTS WORTH KNOWING ABOUT

One table, not a rotation. Wander over during the build.

- Capacitive touch. A wire. See bench 04
- Heart rate (PPG). Useless once the hand moves
- Skin conductance (GSR). Relative change only
- Flex sensor. The cheap data glove
- Pressure (FSR). Calibrate every pad
- Your own phone. No soldering at all

TWO RIGS THAT ALREADY EXIST

Built here, by students at roughly your stage.

Moving screen, Arduino Uno

- Breathes when idle. Retreats when touched
- Easing and idle motion do the work. A dozen lines

Instrumented sock, ESP32

- Six force sensors under a foot, batched over Wi-Fi
- Calibrate per sensor. Batch your writes

BUILD ONE SIGNAL

THE BRIEF

30 minutes. Not long enough to be precious about it.

One actuator. Three messages

- Arrived. Something wrong. Finished

Vary two things only

- Rhythm of the pulses
- How strong they are

THE CONSTRAINT THAT MATTERS

The recipient cannot look at the device.

If it only works while someone watches a screen,
you built a visual interface with a motor glued to it.

GO AND TOUCH THINGS

github.com/gust1527/multimodal-actuator-workshop